Junhyun An

Statistics & Mathematics, Seoul National University | anjunhyun@snu.ac.kr | +82-10-2415-7065

https://anjunhyun.github.io/anjunhyun/

Education

Seoul National University, BS in Statistics and Mathematical Sciences — Mar 2021 – Present
Korea Science Academy of KAIST (KSA) — 2018 – 2020

Research Interests

Knot Theory • Combinatorial Topology • Graph Theory • Low-Dimensional Topology

Research Experience

Research & Education (R&E) — May 2019 – Dec 2019
Advisor: Prof. Gyotaek Jin

The Smallest Lattice Cube Containing Nontrivial Band Knots

- Proved that every band knot in a $3 \times 3 \times 3$ lattice cube is ambient isotopic to the unknot.
- Established the existence of a Möbius strip in $2 \times 2 \times 2$ and a trefoil knot in $4 \times 4 \times 4$ through computational enumeration.

Independent Research — May 2020 – Dec 2020, Feb 2023 – Sep 2023

Computational Complexity and Approximation of the Rural Postman Problem in Euclidean Graphs

- Proved NP-hardness by showing that Euclidean TSP is a special case.
- Developed a polynomial-time upper bound using minimum matching, doubled edges, and a minimum spanning tree.

Undergraduate Research Internship — Dec 2024 – Dec 2025
Advisor: Prof. Jisu Kim

Periodicity Testing for Time Series

- Co-developed a topological method for periodicity testing using one-dimensional homological features.
- Proved asymptotically valid confidence bounds for one-dimensional homology features.

Random Graph Models for Knot Generation

- Developed computational methods to study knot distributions in random graph embeddings.

Persistent Homology of Subsampled Paley Graphs

- Observed relationships between persistent homology and Gauss sum magnitudes, including differences between prime and prime-power fields.

Independent Research — Aug 2026 – Present

Rank-Weighted Coloring Polynomial and the Partial-Dual Genus Polynomial

- Introduced a rank-weighted coloring polynomial for finite simple graphs whose specializations recover the chromatic polynomial and Cheng's partial-dual genus polynomial, and whose binomial-basis coefficients recover Cheng's higher state sums.
- Proved a graph four-term relation for the polynomial and derived an explicit primitive-projection formula for Cheng's partial-dual genus polynomial.

Academic Activities

Knot Theory Seminar — Organizer & Presenter — Mar 2026 – Jun 2026

- Designed and coordinated a nine-lecture seminar, selecting all topics and organizing the progression from classical knot invariants to combinatorial and 4D knot theory.
- Prepared and delivered several lectures and supervised the remaining presentations, reviewing lecture slides and refining mathematical content with two other presenters.
- Designed all seminar exercises and led solution sessions or provided written solutions. *Lecture materials*

Publications

[1] D. Park, **J. An**, T. Kim, and J. Kim. "Topological Periodicity Test (TopPT) via Confidence Bound of Time-Delay Embeddings" *Advances in Neural Information Processing Systems (NeurIPS)*, 2026. **Accepted.** Co-first author.